IDENTIFYING AND IMPROVING TELCO CUSTOMER CHURN

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DSC 630 Predictive Analytics

March 8, 2026



AGENDA

- Introduction of the problem
- Data overview
- Methods of analysis
- Results and findings
- Conclusion



INTRODUCTION

In subscription-based industries, customer churn poses a persistent challenge due to high competition, low switching costs, and growing value-added services.

Predictive analytics provides an opportunity to anticipate churn behavior before it occurs, enabling organizations to take proactive steps to retain customers.

Predicting human behavior is not a simple task, however this proposal outlines a predictive analytics project focused on customer churn using the IBM Telco Customer Churn dataset.



GOAL

This effort seeks to demonstrate how predictive analytics can be translated into actionable business recommendations that support customer retention efforts.

1. Which customer, contract, and service attributes are most strongly associated with churn in this dataset?

2. How well can a classification model distinguish churners from non-churners, especially in terms of recall for churners?

3. How might the model's insights inform concrete retention strategies?



DATA OVERVIEW

	CustomerID	Count	Country	State	City	Zip Code	Lat Long	Latitude	Longitude	Gender
0	3668-QPYBK	1	United States	California	Los Angeles	90003	33.964131, -118.272783	33.964131	-118.272783	Male
1	9237-HQITU	1	United States	California	Los Angeles	90005	34.059281, -118.30742	34.059281	-118.307420	Female
2	9305-CDSKC	1	United States	California	Los Angeles	90006	34.048013, -118.293953	34.048013	-118.293953	Female
3	7892-POOKP	1	United States	California	Los Angeles	90010	34.062125, -118.315709	34.062125	-118.315709	Female
4	0280-XJGEX	1	United States	California	Los Angeles	90015	34.039224, -118.266293	34.039224	-118.266293	Male

1. CustomerID
2. County
3. Country
4. State
5. City
6. Zip Code
7. Lat Long
8. Latitude
9. Longitude
10. Gender
11. Senior Citizen
12. Partner
13. Dependents
14. Tenure Months
15. Phone Service
16. Multiple Lines
17. Internet Service
18. Online Security
19. Online Backup
20. Device Protection
21. Tech Support
22. Streaming TV
23. Streaming Movies
24. Contract
25. Paperless Billing
26. Payment Method
27. Monthly Charges
28. Total Charges
29. Churn Label
30. Churn Value
31. Churn Score
32. CLTV
33. Churn Reason



METHODS OF ANALYSIS (DATA CLEANING)

Number Encoding

Encoded the Churn Label target variable into a new, binary variable for modeling.

Dropped Columns

Removed unnecessary columns for modeling:

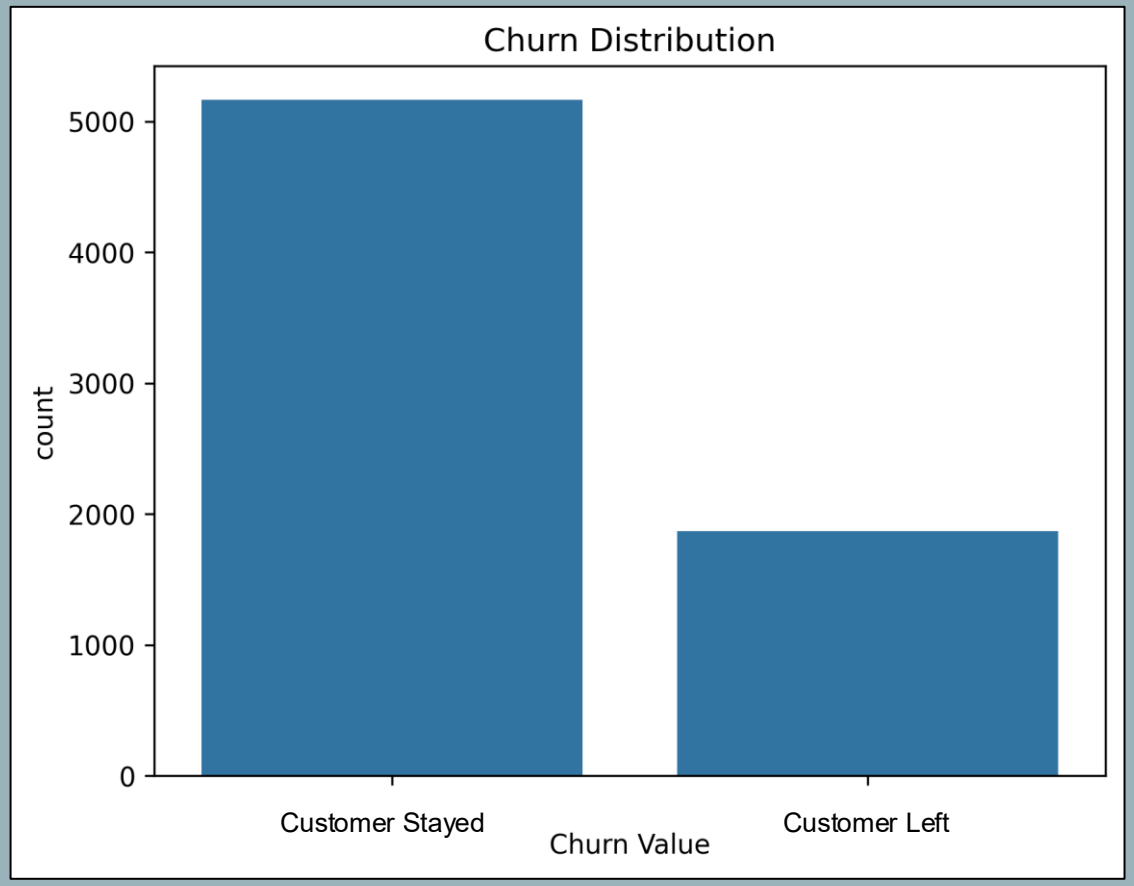
- CustomerID
- Count
- Churn Label
- Churn Score
- CLTV
- Churn Reason

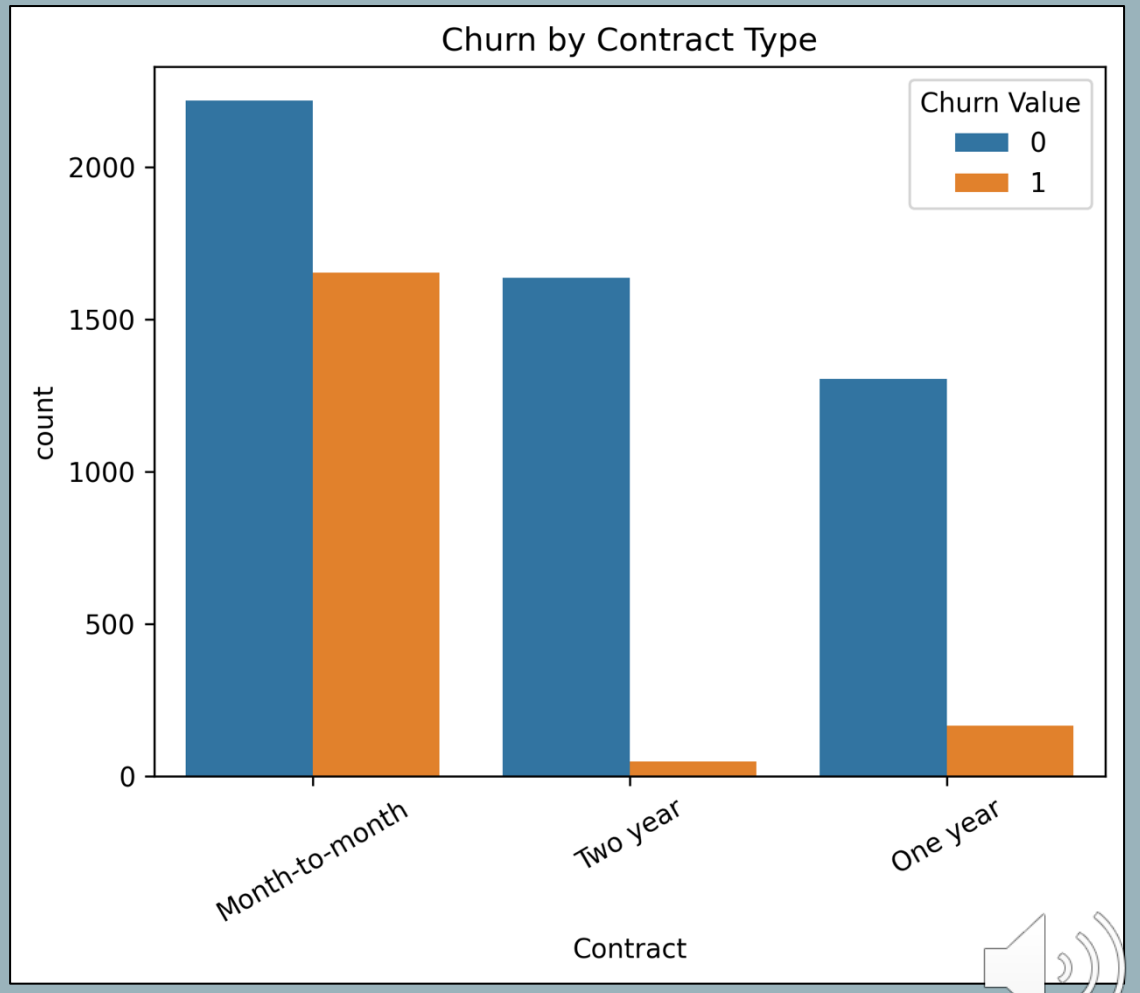
Convert to Numeric

Ensured Total Charges column was numeric.

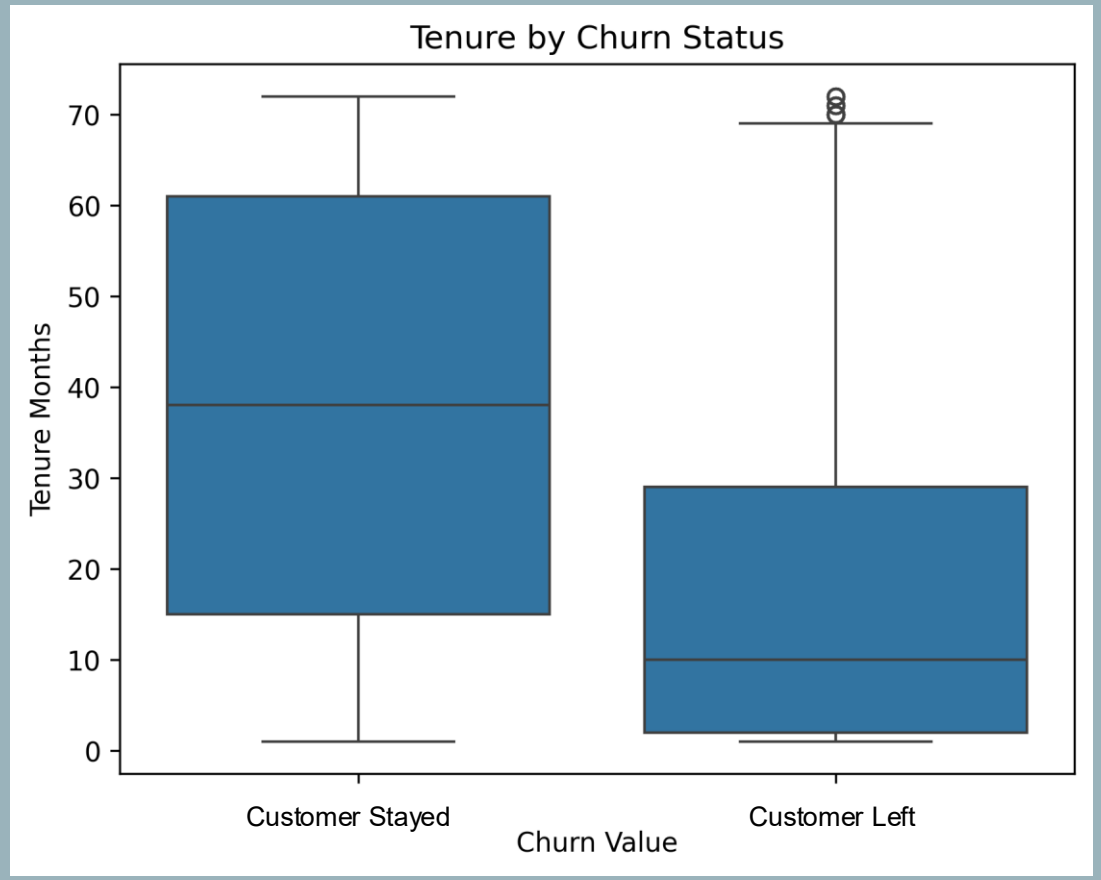


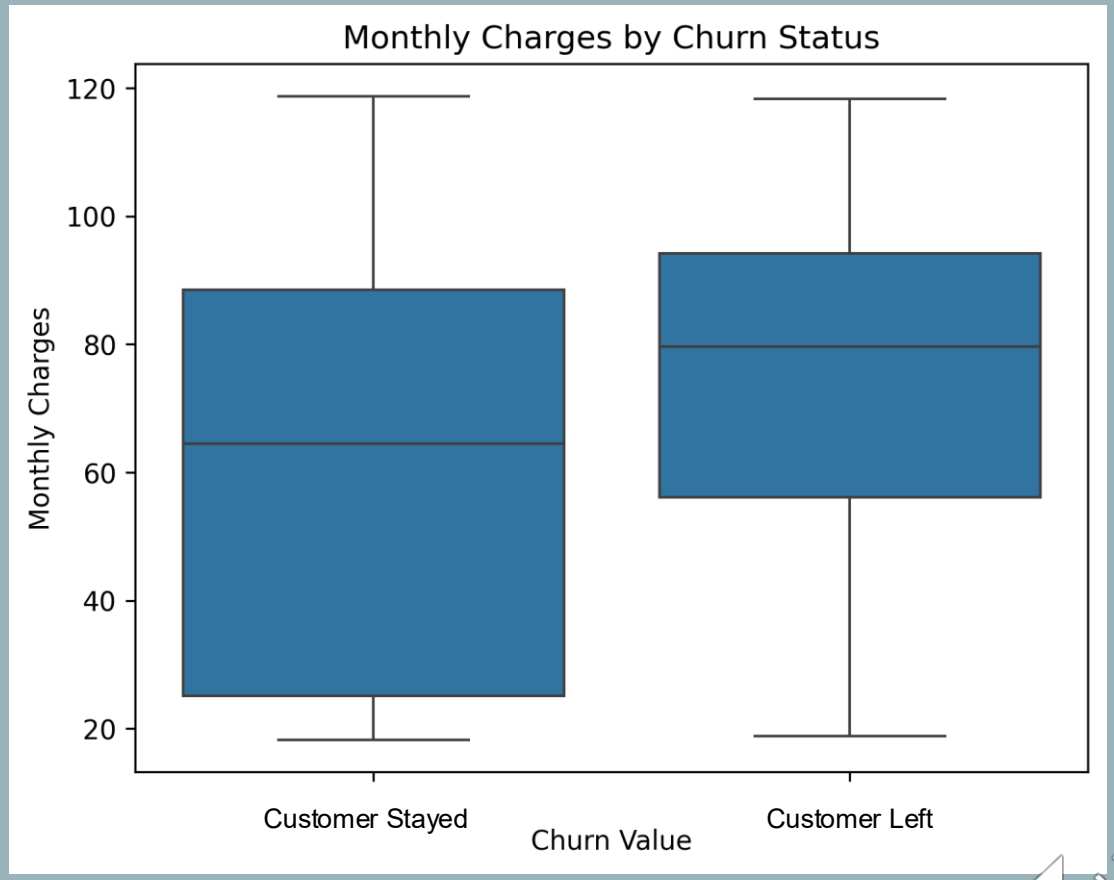
METHODS OF ANALYSIS (EDA)



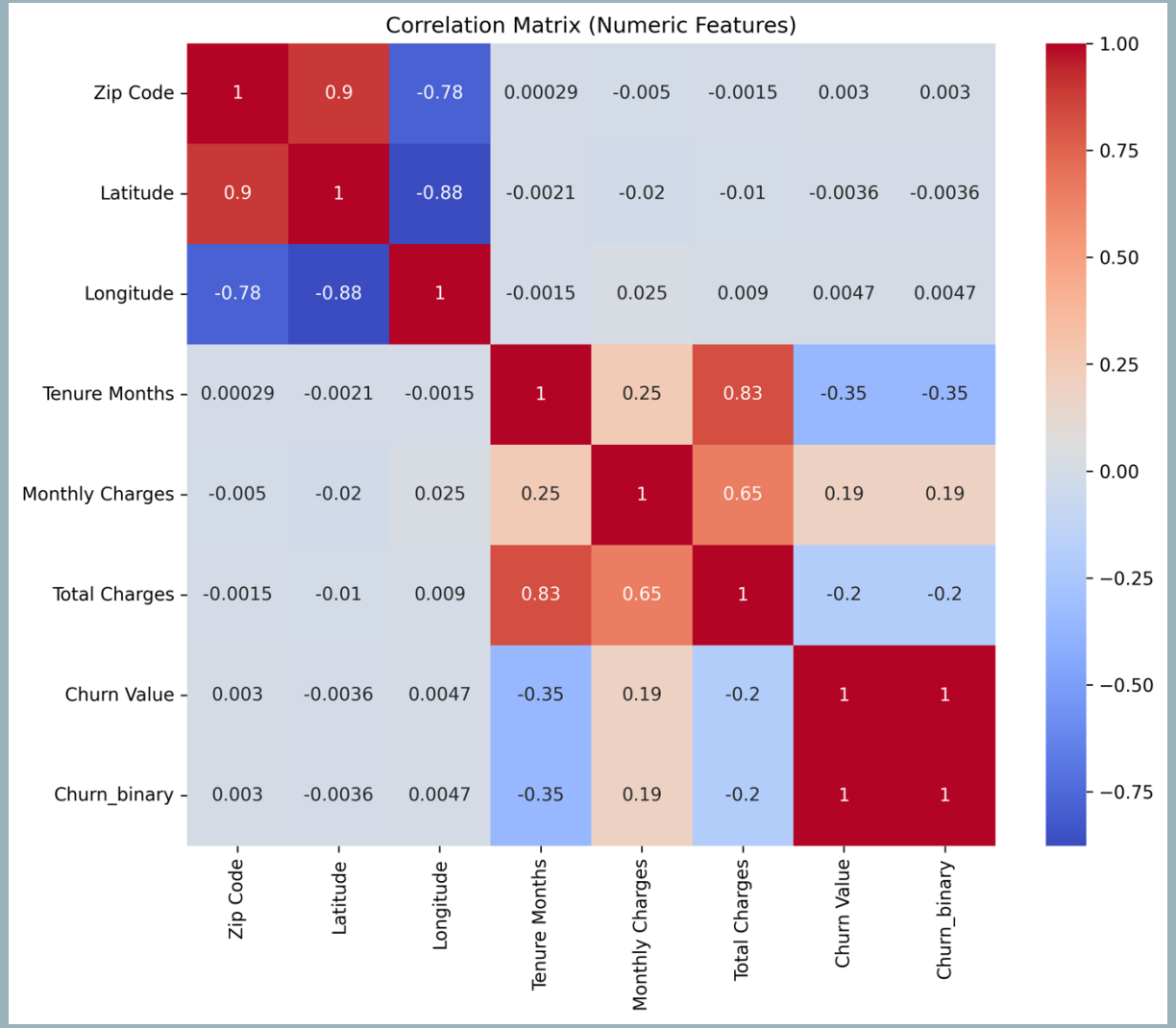


METHODS OF ANALYSIS (EDA)





METHODS OF ANALYSIS (EDA)



RESULTS AND FINDINGS (LOGISTIC REGRESSION)

	precision	recall	f1-score	support
0	0.79	0.81	0.80	1549
1	0.44	0.41	0.42	561
accuracy			0.70	2110
macro avg	0.61	0.61	0.61	2110
weighted avg	0.70	0.70	0.70	2110

Log Regression ROC AUC Score: 0.7041585106370736

Class_weight = ”balanced”

	precision	recall	f1-score	support
0	0.81	0.78	0.79	1549
1	0.44	0.48	0.46	561
accuracy			0.70	2110
macro avg	0.62	0.63	0.63	2110
weighted avg	0.71	0.70	0.70	2110

Log Regression (Balanced) ROC AUC Score: 0.7083875630186343



RESULTS AND FINDINGS (TREES)

Decision Tree

	precision	recall	f1-score	support
0	0.91	0.68	0.78	1549
1	0.48	0.82	0.61	561
accuracy			0.72	2110
macro avg	0.70	0.75	0.69	2110
weighted avg	0.80	0.72	0.73	2110

Tree ROC AUC Score: 0.8269057490946375

Random Forest

	precision	recall	f1-score	support
0	0.84	0.90	0.87	1549
1	0.65	0.51	0.57	561
accuracy			0.80	2110
macro avg	0.74	0.71	0.72	2110
weighted avg	0.79	0.80	0.79	2110

Random Forest ROC AUC Score: 0.8366463787228607



CONCLUSION

Retention strategies should include:

1. Target month-to-month customers within the first year of account creation.
2. Prioritize retention campaigns for high monthly charge customers.
3. Deploy the Random Forest model as it can be used as a risk-ranking system rather than a strict classifier.
4. Adjust thresholds based on cost tolerance.

	Recall (Churn)	Precision (Churn)	ROC-AUC	Accuracy
Logistic Regression	0.41	0.44	0.70	0.70
Logistic (Balanced)	0.48	0.44	0.71	0.70
Decision Tree	0.82	0.48	0.83	0.72
Random Forest	0.51	0.65	0.84	0.80

